MODULE: BIOMOLECULES

Molecules of life:

The study of chemistry of living organisms is known as Biochemistry. The carbon compounds obtained from the living tissue are known as Biomolecules. The different types of biomolecules present in the cell are collectively called cellular pool. The compounds of carbon are central to life on this planet. Carbon compounds include DNA (the giant helical molecules that contain all of our genetic information), RNA (conduit of gene expression), enzymes (that catalyse all of the biochemical reactions in our body). They also include amino acids, proteins, carbohydrate, nucleotide and lipids.

Carbohydrates:

They are hydrates of carbon, containing C, H and O, with the empirical formula $C_x(H_2O)_y$. Based on their size, carbohydrate can be of the following types:

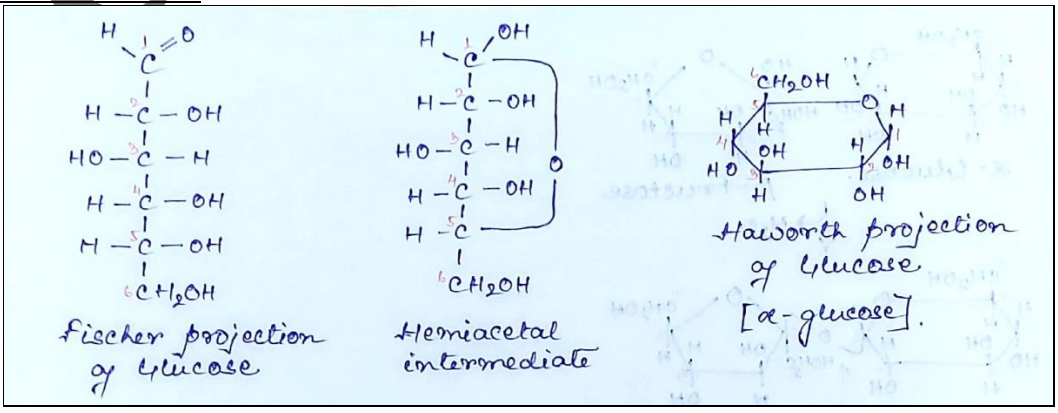
- Monosaccharides: Single unit.
- Oligosaccharides: 2 to 10 units condensed.
- Polysaccharides: More than 10 units polymerised.

Monosaccharides:
They are the basic units of carbohydrates. They are the simplest form of sugar and are usually colourless, water-soluble, crystalline solids. Some monosaccharides have a sweet taste. Some major monosaccharides are Glucose, Galactose and Fructose. Monosaccharides are the building blocks of oligosaccharides and polysaccharides.

Monosaccharides can be classified chiefly by three basis. They are:

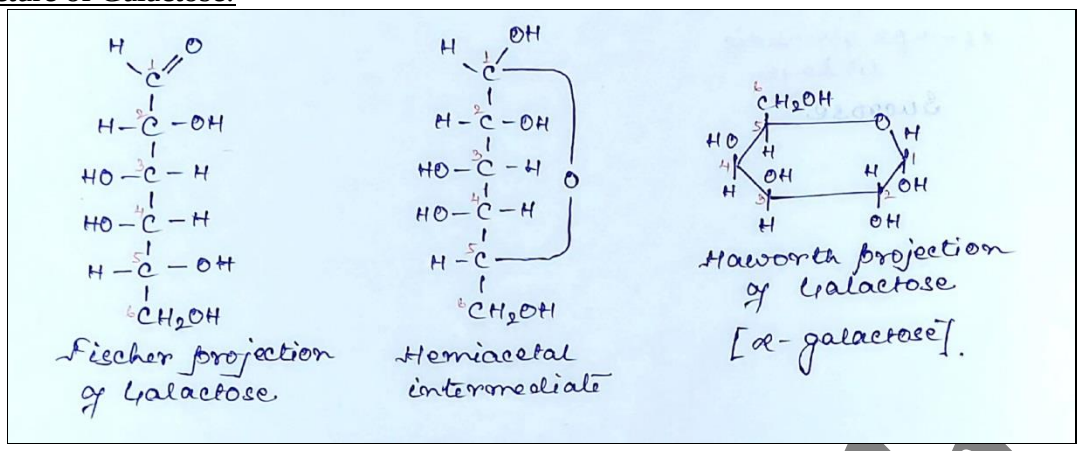
- On the basis of the number of carbon atoms they contain. They include Triose (3C), Tetrose (4C), Pentose (5C), Hexose (6C) and Heptose (7C).
- On the basis of the type of carbonyl group they contain. They include Aldose (aldehyde group containing) and Ketose (ketone group containing).
- On the basis of the attachment of the hydroxyl (-OH) group to the anomeric carbon. Anomeric carbon is the carbon of the carbonyl group which participates in the hemiacetal or hemiketal formation. If the –OH group attached to the anomeric carbon is present below the plane then it is α-anomer, however, if it is present above the plane then it is β-anomer.

Structure of Glucose:

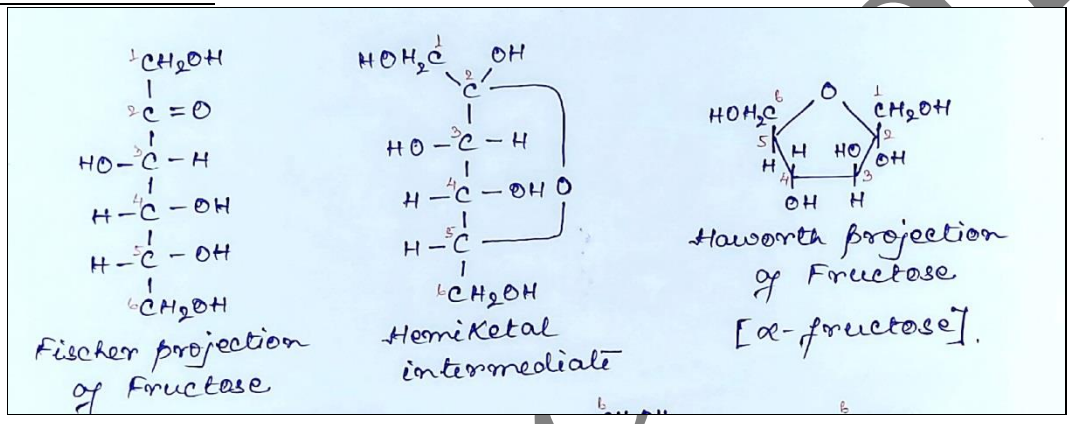


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Structure of Galactose:



Structure of Fructose:



Functions of monosaccharide:

- Monosaccharides like glucose acts as the chief fuel for the driving of cellular activities.
- Monosaccharides largely form component of RNA, DNA, coenzymes (NAD,FAD), energy carriers (ATP), etc.
- They serve as the building blocks for the formation of oligosaccharides and polysaccharides.
- Monosaccharides are used for the formation of different other compounds, for example, glucose is used for the formation of fats, chitin, amino acids, etc.

Disaccharides:

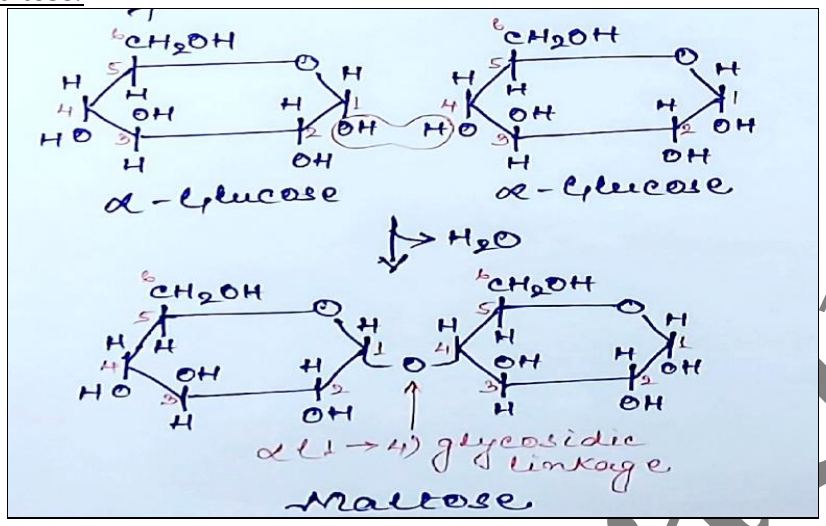
When to monosaccharides undergo a condensation reaction which involves the elimination of a water molecule, from the functional groups only, a disaccharide gets formed. The two monosaccharides are bonded via a dehydration reaction that leads to the loss of a molecule of water and formation of a glycosidic bond. Depending on the monosaccharide constituents, disaccharides are sometimes crystalline, sometimes water-soluble and sometimes sweet-tasting and sticky-feeling.

There are two types of disaccharides:

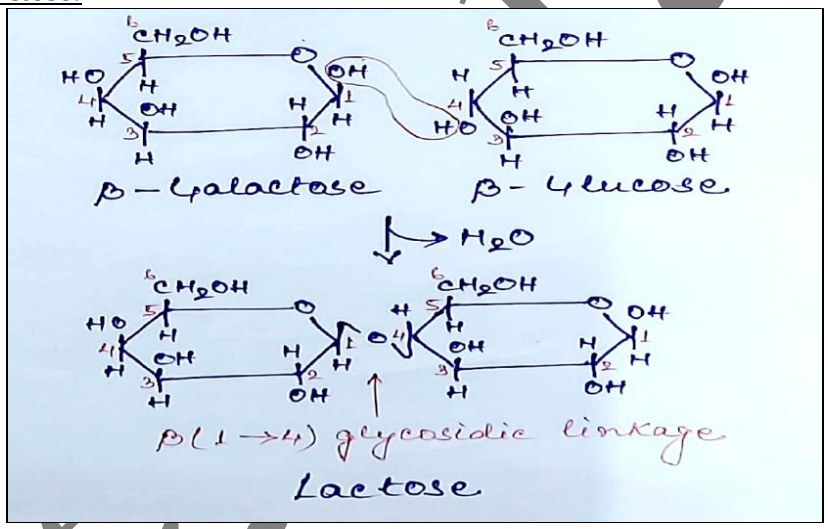
- Reducing disaccharides: In these sugars, at least one of the participating monosaccharide has a free hemiacetal or hemiketal unit, which is capable of reducing. For example, Maltose, Lactose.

- Non-reducing disaccharides: in these sugars, none of the participating monosaccharide has a free hemiacetal or hemiketal unit and are thus incapable of reducing. For example, Fructose.

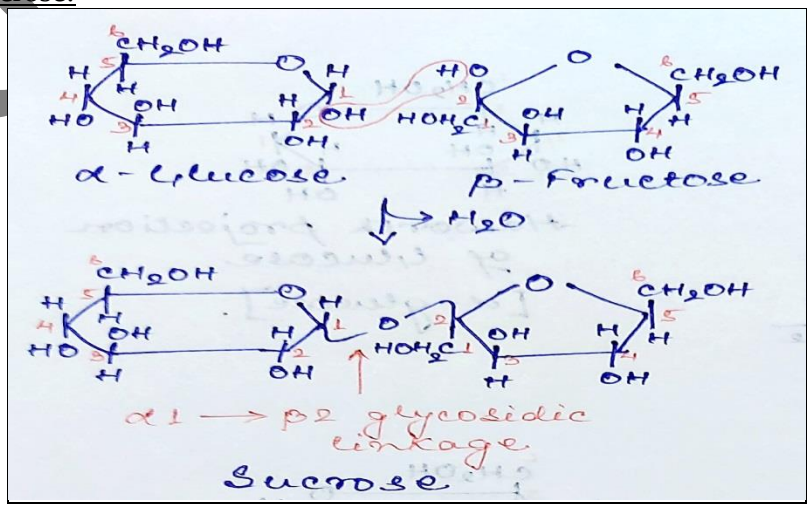
Structure of Maltose:



Structure of Lactose:



Structure of Sucrose:



Functions of disaccharide:

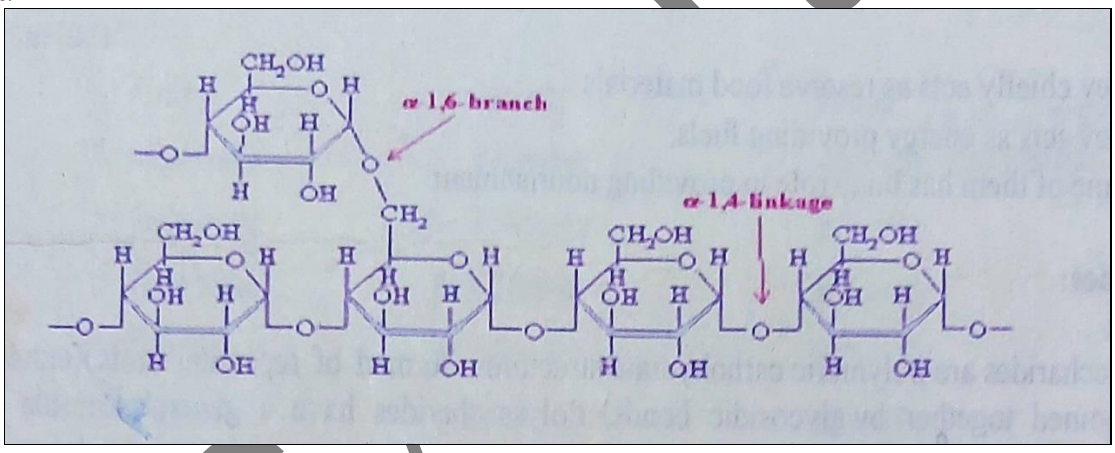
- They chiefly act as reserve food materials.
- They act as energy providing fuels.
- Some of them has huge role in providing nourishment.

Polysaccharides:

They are polymeric carbohydrate structures, formed of repeating units, joined together by glycosidic bonds. These structures are often linear, but may contain various degrees of branching. Depending on the structure, these macromolecules can have distinct properties from their monosaccharide building blocks. They may be amorphous or even insoluble in water.

Starch:

It is a branched, nutrient, storage homopolysaccharide of plant cells. It is merely soluble in water and exists in the cytoplasm as granules. Starch molecules exist in a continuous spectrum of sizes, with the largest molecules containing many thousands of glucose or glucosyl units. Each starch molecule consists of very long, branched chains of α-glucose molecules. The glycosidic bonds are established between C1 and C4 of glucose via α(1→4) glycosidic linkages, except at the branch points, which involve linkages between C1 and C6 of glucose via α(1→6) glycosidic linkages. Branching occurs after every 24 – 30 glucose unit.

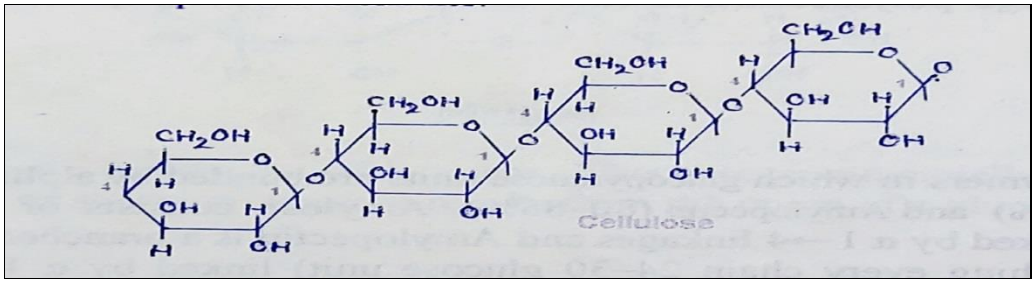


Glycogen:

It is a branched, nutrient, storage homopolysaccharide of all animal cells. It is more water soluble than starch and exists in the cytoplasm as tiny granules. Glycogen molecules exist in a continuous spectrum of sizes, with the largest molecules containing many thousands of glucose or glucosyl units. Each glycogen molecule consists of long, profusely branched chains of α-glucose molecules. The glycosidic bonds are established between C1 and C4 of glucose via α(1→4) glycosidic linkages, except at the branch points, which involve linkages between C1 and C6 of glucose via α(1→6) glycosidic linkages. Branching occurs after every 8 – 10 glucose unit.

Cellulose:

It is a linear, structural homopolysaccharide of plant cells. It is perfectly insoluble soluble in water and forms the cell wall of all plant cells. Each cellulose molecule consists of very long chains of β-glucose molecules. The glycosidic bonds are established between C1 and C4 of glucose via β(1→4) glycosidic linkages.



Functions of polysaccharides:

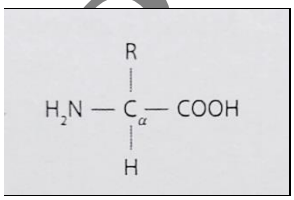
- Polysaccharides like starch, glycogen act as ideal storage product.
- Polysaccharides like starch and glycogen are chief fuel substance and thus serve as the primary source of energy.
- Polysaccharides like cellulose act as chief structural compound as well.

Amino acids and Proteins:

Amino acids:

They are the monomers of proteins. Each amino acid has an amino group ($-NH_2$), an acidic carboxyl group (-COOH) and a side chain (-R) group. Twenty amino acids have been found to be present in proteins. All amino acids are chiral except glycine.

General structure of amino acid:

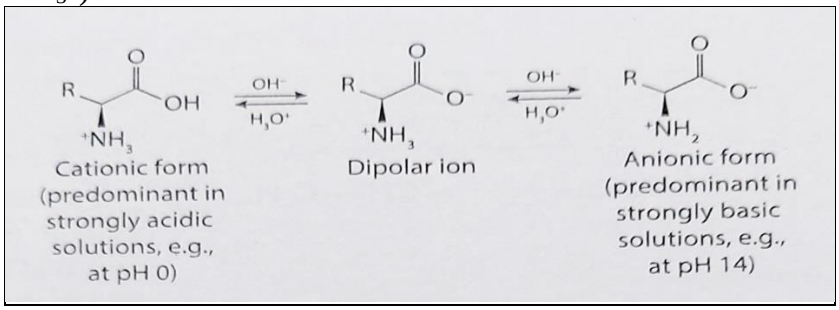


Classification of amino acids:

- Amino acids with non-polar R groups: Glycine, Alanine, Valine, Leucine, Isoleucine, Methionine, Proline, Phenylalanine and Tryptophan.
- Amino acids with uncharged polar R groups: Serine, Threonine, Asparagine, Glutamine, Tyrosine and Cysteine.
- Amino acids with positively charged polar R groups: Lysine, Arginine and Histidine.
- Amino acids with negatively charged polar R groups: Aspartic acid and Glutamic acid.

Zwitterionic form of amino acids:

The amino acids contain both acidic (-COOH) and basic ($-NH_2$) group. These groups are ionisable. The amino acids occur as dipolar ions or zwitterions in dry state. In this form, the carboxyl group is present as carboxylate ion ($-COO^-$) and the amino group is present as an aminium ion ($-NH_3^+$).



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Essential and non-essential amino acids:

The amino acids that can be synthesized by the body are non-essential amino acids. Those that cannot be synthesized by the body need to be supplemented in the diets. Such amino acids are known as essential amino acids. Valine, leucine, isoleucine, phenylalanine, tryptophan, histidine, threonine, methionine and lysine are essential amino acids.

Proteins:

Proteins, also known as polypeptides are organic compounds made of amino acids arranged in a linear chain and folded into a globular form. The amino acids in a polymer are joined together by the peptide bonds between the carboxyl and amino groups of adjacent amino acid residues. The sequence of amino acids in a protein is defined by the sequence of a gene, which is encoded in the genetic code.

Nucleotides and Nucleic acids (DNA/RNA):

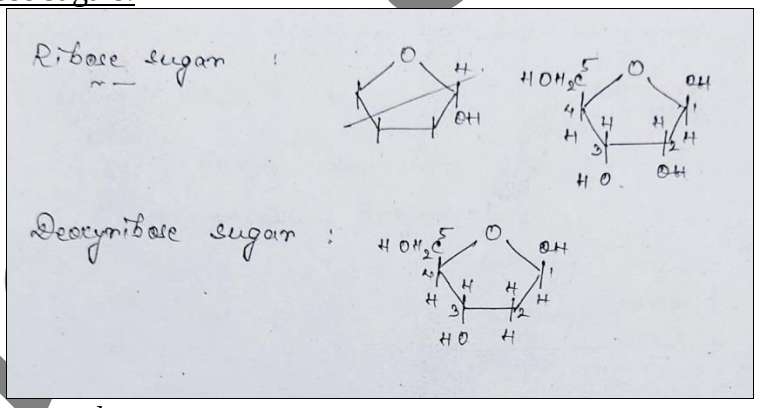
Nucleotides:

They are molecules that, when joined together, make up the structural units of RNA and DNA. A nucleotide is composed of a nitrogenous base (either purine or pyrimidine), a five carbon sugar (either ribose or deoxyribose) and a phosphate group. Together, the nitrogenous base and sugar comprise a nucleoside. The phosphate group forms bond with the 5th carbon and completes the nucleotide formation.

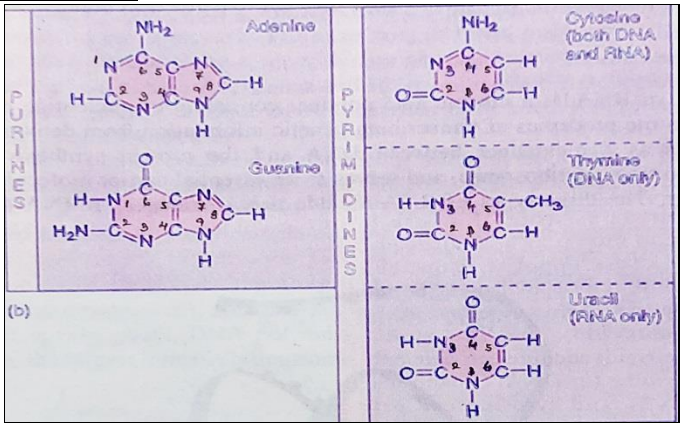
Ribonucleotides are nucleotides where the sugar is ribose, nitrogenous base if purine is either adenine or guanine, if pyrimidine is either cytosine or uracil.

Deoxy ribonucleotides are nucleotides where the sugar is deoxy ribose, nitrogenous base if purine, is either adenine or guanine, if pyrimidine, is either cytosine or thymine.

Structure of pentose sugars:

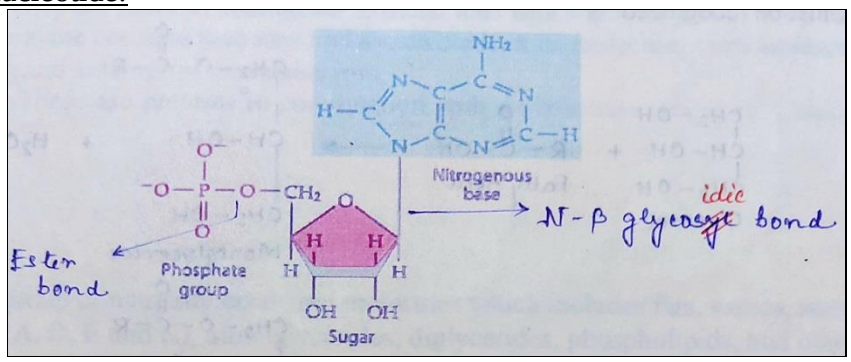


Structure of nitrogenous bases:



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Structure of nucleotide:



Functions of nucleotide:

- They are the structural unit of nucleic acids.
- They play central role in metabolism.
- They serve as sources of chemical energy (ATP, GTP, etc.)
- They participate in cellular signalling (cAMP, cGMP, etc.)
- They cat as important cofactor of enzymatic reactions (CoA, FAD, FMN, NADP, etc.)

Nucleic acids:

A nucleic acid is a macromolecule composed of chains of monomeric nucleotides. In biochemistry these molecules carry genetic information or form structures within cells. The most common nucleic acids are Deoxyribonucleic acid (DNA) and Ribonucleic acid (RNA).

Ribonucleic acid:

It is a nucleic acid polymer consisting of ribonucleotide monomers, which plays several important roles in the process of transcribing genetic information from deoxyribonucleic acid into proteins. There are three types of RNA – mRNA, rRNA and tRNA. mRNA acts as a messenger between DNA and the protein synthesis complexes known as ribosomes, rRNA forms vital portions of ribosomes and tRNA serves as an essential carrier molecule for amino acids to be used in protein synthesis.

Deoxyribonucleic acid:

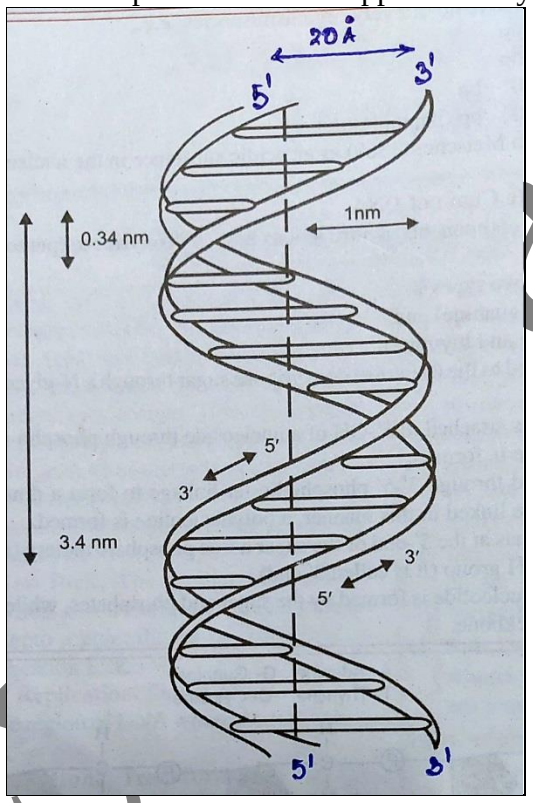
It is a nucleic acid that contains the genetic instructions used in the development and functioning of all known living organisms. The main role of DNA molecule is the long-term storage of information and is often compared to a set of blueprints, since it contains the instructions needed to construct other components of cells, such as RNA molecules and proteins. DNA is made up of four types of nucleotides, containing different nitrogenous bases, the pyrimidines (cytosine and thymine) and the purines (adenine and guanine). The nucleotides are attached to each other in a chain by bonds between their sugar and phosphate groups, forming a sugar-phosphate backbone. Two of these chains are held together by hydrogen bonding between complementary bases (adenine pairing with thymine via two hydrogen bonds and guanine pairing with cytosine via three hydrogen bonds). The chains coil around each other forming the DNA double helix.

In 1953, James Watson and Francis Crick based on the X-ray diffraction data produced by Maurice Wilkins and Rosalind Franklin proposed a very simple but famous double helix model for the structure of DNA. The proposition of the model was also based on the observation of Erwin Chargaff that for a double stranded DNA; the ratios between purine and

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pyrimidine are constant and equal one. Salient features of the Double-helix structure of DNA are as follows:

- It is made up of two polynucleotide chains, where the backbone is constituted by sugar-phosphate, and the bases projects inside.
- The two chains have anti-parallel polarity. It means, if one chain has the polarity 5' → 3', the other has 3' → 5'.
- The bases in two strands are paired through hydrogen bond forming base pairs (bp).
- The DNA chain is approximately 20 Å wide.
- The two chains are coiled in a right-handed fashion.
- The pitch (on complete turn of the DNA) of the helix is 3.4 nm.
- There are roughly 10 bp in each turn.
- The distance between a base pair in a helix is approximately equal to 0.34 nm.



Lipids:

Lipids make up 18 – 25% of adult body mass, and contain carbon, hydrogen and oxygen. They are insoluble in polar solvents such as water, hence hydrophobic in nature. Fatty acids are the simplest and major constituents of all lipids. Fatty acids are basically monocarboxylic acid containing short or long chain hydrocarbon molecules. A fatty acid consists of a carboxyl group and an R group. The R group can be 5 carbons to 36 carbons long, though in naturally occurring fatty acids the R group are 12 carbons to 18 carbons long.

Types of fatty acids:

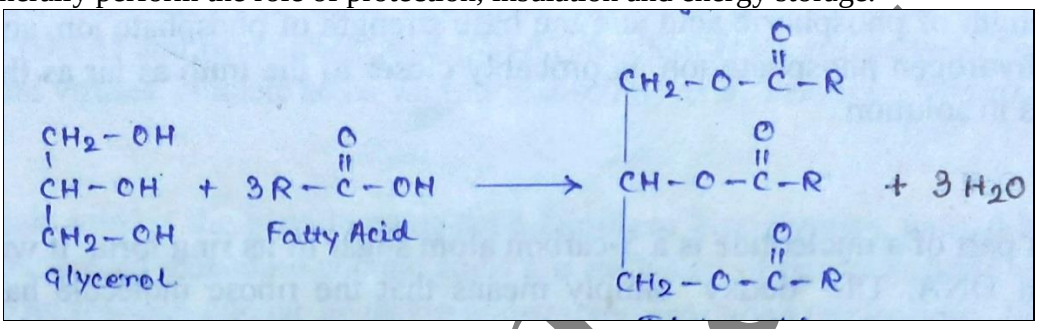
- Saturated fatty acids: They contain only one single covalent bond (C – C) between the carbon atoms of the hydrocarbon chain. For example, stearic acid, palmitic acid, etc.
- Unsaturated fatty acids: They contain one or more double covalent bonds (C = C) between the carbon atoms of the hydrocarbon chain. For example, oleic acid, linoleic acid, linolenic acid, etc.

Essential and non-essential fatty acids:

Both plants and animals have the biosynthetic machinery to manufacture fatty acids, however, animals are unable to manufacture linoleic, linolenic and arachidonic acids. They must be obtained from foods or supplements and are called essential fatty acids. However the fatty acids that can be synthesized within the body itself are called non-essential fatty acids.

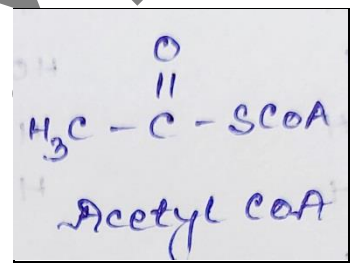
Classification of lipids:

- Triglyceride (Fats and oils): They are simple lipid, formed by the esterification of a single glycerol molecule with three fatty acid molecules attached to its three carbons. A fat is a triglyceride that is solid at room temperature, they are mostly saturated. Oil is triglyceride that is liquid at room temperature, they are mostly unsaturated. They generally perform the role of protection, insulation and energy storage.



- Waxes: They are simple lipid, esters of long chain fatty acids and long chain alcohols. They form protective coatings on the skin, fur and feathers of animals and on the leaves and fruits of plants.
- Phospholipids: They are compound lipid. They have a glycerol backbone, to which two fatty acid chains are attached to the first two carbons and a phosphate group attached to the third carbon. They are the major lipid component of cell membrane.
- Steroids: They are derived lipid. They are derivative of cyclopentanoperhydrophenanthrene ring. They are major component of cholesterol, Vitamin D, certain hormones, etc.

Two carbon units (Acetyl CoA):



Acetyl coenzyme A (Acetyl CoA) can be regarded as a focal point in metabolism. It is a common high energy substance produced by the oxidation of fatty acids, amino acids and pyruvate. It is utilised in many ways and referred to as the 'Building stone' of fatty acid synthesis. Its different utilisation is:

- Complete oxidation of glucose by the TCA (Tricarboxylic Acid) cycle.
- Fatty acid synthesis.
- Acetylation of cholines and aromatic amines.
- Synthesis of cholesterol, steroid hormones, bile acids, etc.

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